Design Report: "To Beam or Not to Beam" Bridge

1.0 Introduction

For a three-span railroad bridge to be constructed over a valley, we have detailedly designed, calculated and built one section of the bridge. The model, made solely out of matboard and contact cement, is to be subjected to two tests: supporting a 400N train moving along it, as well as two point loads at two specified points. The following brief discusses the requirements of this project and highlights the key design decisions that have been made. Our bridge has two cross sections to better resist the bending moments; its decks are simply bent rather than glued onto the flange; there are additional layers on the top and the bottom of the bridge in different sections. The reasoning behind our design takes into account ease of construction, limitation of materials, and the overall goal to achieve maximum loading. All design decisions are thoroughly analyzed and justified in this report.

2.0 Engineering Design requirements

2.1 Constraints

1. The design must be constructed with the given materials and resources
2. The design must fit the required dimensions
3. The design must meet the primary objective that is set

2.2 Criteria and Metrics

1. The amount of material used should be as little as possible.
2. The design should be aesthetically appealing - measured qualitatively
3. The design should meet the calculated load

3.0 Bridge Design Requirements

3.1 Constraints

3.1.1 Train Loading

The bridge must be able to support a 400N train.

3.1.2 Length of the Bridge

The bridge must have a length greater than 1250 mm, but less than 1325mm.

Metric: Length of the bridge.

3.2.1 Materials

The bridge must only use a single 32”x40”x0.05” sheet of matboard and 60 mL of contact cement.

3.2 Criteria

3.2.1 Point Loads

The bridge should be able to support as large a P from the given loading configuration as possible

Metric: The value of the maximum point load.

3.2.2 Construction

The bridge should be easy to construct and allow for minimal errors to occur during the construction process.

Metric: Number of cuts and amount of glueing needed to construct the bridge.

3.2.3 Train Passage

The bridge must provide unhindered passage for the train.

Metric: Number of ways the train’s movements impeded.

4.0 Statement of Bridge Design Philosophy

4.1 Design for Safety

“Safety is always our priority.”

4.2 Design for Manufacturing

“A simple, elegant design, that minimizes human error, so that the theory represents reality.”

4.3 Design for Practicality

“Getting the job done efficiently with the materials at hand.”

5.0 Key Design Decisions

5.1 Having Two Cross-Sections:

As seen in Appendix A, the bending moment diagram for the point loads consists of two sections: one with positive moment, labelled A, and one with negative moment, labelled B. As section A is purely a pi-shaped cross section with no cover at the bottom, the centroid is very high, resulting in a good performance in resisting compression on the top. The consideration of compression is especially important, because the failing compressive stress is only 6 MPa, much lower than the maximum tensile stress of 30 MPa.

However, due to a negative moment in region B, compressive stresses are on the bottom of the cross section, which, along with a high centroid, leads to a significantly high compressive stress. Therefore, it is necessary to change the design of the cross section for the region with negative moments.

Thus, cross-sections are customized for different regions: cross-section B has a layer of Matboard beneath it with an attempt to increase I and lower the centroid. Such a design reduces the compressive stresses at the bottom and increases the predicted loading as a result. This design strengthens our bridge.

5.2 Bending the Matboard deck rather than gluing the web:

As seen in the drawing of the bridge, the bottom layer of the deck and the webs of the bridge are one piece of matboard bent into a pi shape. This design avoids the procedure to glue separate pieces of webs onto the flange, which improves the ease of construction and lowers the possibly of a failure mode. As the maximum shear stress of cement glue is less than the maximum shear stress of matboard, gluing the web on will increase the possibility of another failure mode of the bridge, with the top flange being sheared off of the web. Moreover, bending the matboard, rather than gluing it, makes the construction of the bridge easier and minimizes the chance of human error regarding application of cement glue, leading to a more robust design. The cement saved from gluing the web can be utilised in other parts of the bridge.

5.3 Having Two Layers on the Top Deck:

A single layered flange is not strong enough to resist the bending moment. Ideally, we would have more layers on the deck but this is impossible to achieve due to the limited materials. On the other hand, additional deck layers change the centroid of the cross section. After careful consideration over several designs, we found that two layers on the top deck strikes a balance between material use and maximum loading, as having only one layer significantly reduces the loading, while more than two layers uses up more material than necessary.

In our design, both cross section A and cross section B have two layers on the top, ensuring that the train can roll seamlessly through the bridge and simplifying the construction process. If section B was to have a different number of layers on the top as opposed to A, either the top part of cross sections A and B need to be joined together via glue in the regions with high bending moments, or the top layer is only glued on section A, creating a "step" on the track for the passage of the train. Neither of those scenarios are preferable or acceptable, so it's important that section B also has two layers on top.

5.4 Having One Layer on the Bottom of Section B

An additional layer is added to the bottom of section B. If it was not, then section A would have to carry_higher_compressive forces. However, the reason the bottom was chosen to have one layer (as opposed to two like the deck) is because if the bottom contained more layers, then material would need to be taken from elsewhere. If it were taken from the top deck, then section A, as mentioned previously, would not be able to carry too much load, and the bridge would fail at A, negating this improvement. In addition, were the materials taken from the diaphragms, then shear buckling would be the source of failure, thus again negating the point of this improvement.

6.0 Consideration of Design Alternatives

6.1 Single Cross-Section

As mentioned in Section 5.1, a design where the bridge contained a single cross section like cross-section A was considered before the current design was chosen. However, with this design, a region of negative moments in the point loading would result in very high compressive stresses at the bottom, yielding a small maximum loading. As we converged in the design process, this alternative was determined to be infeasible.

It was also considered to extend the cross section B to the entire bridge, as section B of the bridge could resist the highest moment during point loading. However, this alternative design would have used more materials than necessary as cross-section A was not responsible for many critical failure modes; additionally, it would have reduced the amount of material available for diaphragms, eventually decreasing the maximum load even further.

6.2 Gluing the web to the flange

As mentioned in 5.2, we chose to bend the matboard rather than glue the flange to the web. The alternative would be to glue the web onto the flange. This alternative would require gluing separate pieces of webs onto the flange, which made construction process more challenging. Gluing the web on would also increase the possibility of glue shear stress failure. It would also increase the chances of human errors during the construction. Based on the reasons above, in the convergence process of our design, we made the decision to bend the matboard rather than glue the web onto the flange.

6.3 Different choices of number of layers on the top and bottom

In sections 5.3 and 5.4, it was mentioned why the specific choice of layers were chosen in the current design. Different numbers of layers were considered; however, as mentioned in these sections, they either did not provide enough strength to the actual design, or they required too much material, which would have resulted in failures in other parts of the design.

7.0 Basis for Key Design Decisions

7.1 Ease of Construction

As many good designs might fail due to poor construction, we minimized the amount of cuts and gluing that we were required to do as a basis for our choice of number of layers on the cross-sections of the bridge, in order to prevent as many human errors as possible during construction.

7.2 Material Use

There were several designs that would have increased the load the bridge could take, such as increased height of the bridge or width. However, due to the limitation of materials, this would have resulted in either a lack of matboard available for construction, or less diaphragms available for the bridge, which would have resulted in even lower strength, as the bridge would be susceptible to shear buckling failure.

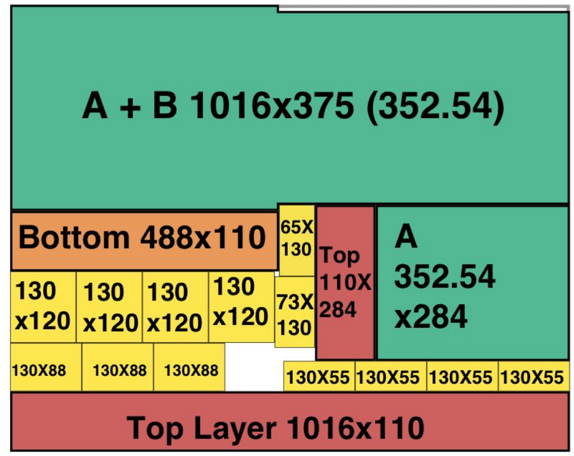
7.3 Maximum Loading Possible

After the other two constraints were considered, we designed the bridge to allow for the largest loading possible. After trying different arrangements that did fit the constraints, such as only 1 very tall cross-section, akin to cross-section A, or having less diaphragms, all of these resulted in failures that arose in other areas, such as shear buckling or failure due to negative moment. As such, the main optimization method we went through was trying to improve the maximum point load the bridge could take before failing.

8.0 Design Evaluation

8.1 Material Used:

The picture on the right represents the allocation of the matboard used for the design. As per the requirements and constraints regarding material usage, the design fits within the single sheet of matboard allowed. In addition, the required measurements of the bridge are met, meaning the design in question is feasible.



8,2 Main objective: Train

The current design is able to support the 400 N train, as per Appendix A, and does not have any “steps” on the deck, so that the train may pass unhindered, thus meeting the main requirement of this project. In addition, there are no obstacles in its path, meeting yet another requirement.

8.3 Ease of Construction

Due to how the bridge is bent, rather than cut or glued, the construction process is made easier, thus meeting the criteria regarding ease of construction. The amount of gluing and cutting is minimized, and it is avoided in this scenario, to prevent the design from failing due to poor construction.

8.4 Maximum Point Load

The maximum point load the bridge may take is 870N. Due to the amount of both positive and negative moment, and the limitation of materials, this is a respectable loading. While not very high, this design minimizes the possibility of construction error, as mentioned in 8.3, thus making sure this loading may more accurately represent reality.

As a result of the above reasoning, we believe our design to meet the constraints and criteria, and is a safe, practical, and easy to build design.

Appendix B: Time Log

Task	Completed By	Time Spent	Signatures
Planning	Raghav, Lisa, Lunjun	Nov 23 9:30 am - 10:50 am	
Point Load and Deflection Calculations	Raghav	Nov 23 8:00pm - 10:00 pm Nov 25 4:00pm - 11:00 pm	
Cross Section Calculations	Raghav	Nov 23 8:00pm-8:30pm Nov 25 4:00-5:00	
Double checking Train Load	Lisa	Nov 26 3:30 pm - 4:30 pm	
Engineering Drawings	Lisa	Nov 25 10 pm - 11pm	
Report	Raghav, Lisa, Lunjun	Nov 25 11:30pm - 12:30pm Nov 26	
Tracing Matboard	Lisa	Nov 26 4:40 pm - 6:20pm	
Train Load calculations	Lunjun	Nov 25 3:00 pm to 5:00 pm Nov 26 10 am to 2 pm	
Double checking Point Load	Lunjun	Nov 26 2 pm to 4 pm	